Shape and Space 5 — Transformations: translation, reflection, rotation, enlargement

Squared paper (and tracing paper for rotations) needed. Label every image. To *describe* a transformation, give all the details: vector / mirror line / angle, direction and centre / scale factor and centre. Marks in brackets. ★ = harder question.

1 Draw axes from -6 to 6. Draw triangle J with vertices $(-2, 3)$, $(2, 3)$ and $(1, 5)$. [8]

- **a** Translate J by the vector $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$. Label the image K.
- **b** Reflect K in the line $y = -x$. Label the image L.
- **c** Reflect L in the line $x = -3$. Label the image M.
- **d** Describe fully the single transformation that maps M onto J.

2 Draw axes from -6 to 6. Draw triangle A with vertices $(-2, -1)$, $(-1, -1)$ and $(-1, 1)$. [12]

- **a** Rotate A $90°$ clockwise about $(1, 2)$. Label the image B.
- **b** Enlarge B by scale factor 2, centre $(-4, 6)$. Label the image C.
- **c** Translate C by $\begin{pmatrix} 2 \\ -6 \end{pmatrix}$. Label the image D.
- **d** Rotate D $90°$ anticlockwise about $(-3, -5)$. Label the image E.
- **e** Describe fully the single transformation that maps E onto A.

3 The point $P(a,\ b)$ is mapped to $Q(10,\ 10)$ by the following transformations *in this order*: reflection in $y = x$; rotation of $90°$ clockwise about O; translation by $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$; enlargement of scale factor 2 about O. Find a and b. [5]

Total: 25 marks

Describe fully: translation → vector · reflection → mirror line equation · rotation → angle, direction, centre · enlargement → scale factor, centre. Useful rules: $y = x$: $(x, y) \to (y, x)$; $y = -x$: $(x, y) \to (-y, -x)$; $90°$ clockwise about O: $(x, y) \to (y, -x)$; $90°$ anticlockwise about O: $(x, y) \to (-y, x)$.